

User Association based Load Balancing using Reinforcement Learning in 5G Heterogeneous Networks

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Abstract

The deployment of Heterogeneous Networks (HetNets), which include macro and micro Base Stations (BSs), has emerged as an effective solution to address the challenges faced by 5G communication networks. HetNets leverage the strengths of various technologies to enhance coverage, capacity, Quality of Service (QoS), cost efficiency, and adaptability. Efficient methods for associating User Equipment (UE) with BSs in HetNets are necessary to maximize overall network performance. This process, known as user association-based load balancing (UALB), is crucial due to the increasing number of devices requiring connectivity using limited available spectrum and BSs. This study considers a three-tier down-link HetNet consisting of Macro BSs (MBS), Pico BSs (PBS), and Femto BSs (FBS), operating under a log-distance channel model in both rural and urban areas, in accordance with 3GPP norms. The signal model for the proposed Het-Net is devised to determine the Signal-to-Interference-plus-Noise Ratio (SINR) and transmission rates for each user association. Load balancing is then performed using the Deep Q-Learning approach, and the cumulative transmission rate achieved is calculated and tabulated. This approach is compared with the MaxSINR and Q-Learning algorithms. LB (Load Balancing) was performed for different network scenarios by varying the number of UEs and BSs and introducing mobility in the network. The corresponding results were analyzed and

plotted. An extensive survey of the effect of user association LB on improving the cumulative transmission rate of HetNets was conducted successfully, and the variation in performance under different network scenarios was analyzed.

Keywords: UARA, Q-Learning, Reinforcement Learning, HetNets, Load balancing, 5G

1 Introduction

The rising demand for faster data speeds, reduced latency, and improved connectivity has driven the development and adoption of 5G technology, marking a significant advancement in wireless communication systems. The surge in data-intensive applications namely streaming, gaming, and Internet of Things (IoT), coupled with the rapid increase in Internet-connected devices [1], challenges traditional wireless networks to meet the evolving need of users and industries. In response, 5G technology aims to deliver higher data rates, minimal latency, and greater capacity to support the growing number of connected devices. This revolutionary step not only addresses current demands but also paves the way for future technological innovations across various fields.

LTE (Long Term Evolution) and LTE-Advanced focus on enhancing wireless communication by providing uplink data transmission speeds of up to 500 Mbit/s and downlink speeds of up to 1 Gbit/s, with latency reduced to less than 10 milliseconds [2]. They offer bandwidth allocations of 70 MHz for downlink and 40 MHz for uplink, designed to improve data transmission speeds at cell edges, reduce power consumption, and optimize spectrum utilization. In contrast, 5G technology reduces latency to 1 millisecond and increases data transmission speeds by tenfold, ranging from 1 to 10 Gbps. It supports connections among thousands of devices in a confined area, with speeds of up to 500 kilometers per hour, guarantees nearly 100 percent coverage, emphasizes energy efficiency and sustainability, and extends battery life [3]. 5G represents a significant evolution in wireless communication technology, surpassing LTE and LTE-Advanced.

The 5G architecture [4] comprises the Next Generation - Radio Access Network (NG-RAN), Core Network (5GC), and UEs. UEs consist of a Mobile Station and a Universal Subscriber Module (USIM) and communicate wirelessly with Base Stations (BS), or gNBs (Next Generation NodeB) via the NG-RAN. The NG-RAN enhances data speeds, latency, and device density through technologies such as massive MIMO, beamforming, and dynamic spectrum sharing. The 5G architecture is illustrated in Figure 1. It supports network slicing, creating virtualized, isolated networks tailored for specific use cases to optimize service performance. To accommodate high mobility users, the 5G cellular architecture needs to be heterogeneous, incorporating relays, small cells, macrocells, and microcells [5]. HetNets are considered a viable solution to the increasing mobile data traffic [6].

A three-tier 5G HetNet architecture typically includes macro, micro, pico, and femto cell layers and access technologies. Macrocells are usually deployed on rooftops, macrocell towers, or other elevated structures. Picocells are often placed indoors or in

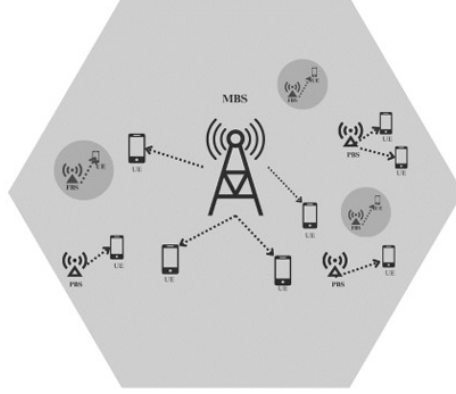


Fig. 1 Architecture of HetNet

areas with high-density requirements, while femtocells are installed in small areas by end users to extend localized coverage and are connected via broadband Internet.

In a 5G wireless network, UEs are connected to the backhaul network via Base Stations (BSs) within a particular cell. Each BS has a limit to the number of UEs it can serve. As the number of UEs rapidly increases, there are instances where the number of UEs in a BS's coverage area exceeds its capacity, causing BS overload. When this occurs, some UEs cannot establish a connection with the overloaded BS. These UEs can instead connect to a neighboring BS with available capacity. This process, known as LB, helps improve the overall capacity and throughput of the network by efficiently distributing UEs among BSs, ensuring optimal spectrum allocation. An effective load-balancing method should ensure that all UEs are connected to a BS while maximizing the network's overall throughput.

A LB constantly monitors the network parameters and detects load imbalances in a network. Following the detection of UEs in congested cells seeking connectivity, an optimization algorithm is executed to identify possible BSs for offloading the extra UEs. After performing LB, the network parameters are monitored again to ensure that the load is balanced between all cells. Finally, the performance indicators were computed and studied to check the efficiency of LB.

LB can be enhanced using machine learning algorithms. One such approach is reinforcement learning, where agents learn to make optimal sequential decisions through trial and error. Agents receive positive and negative numerical rewards for correct and incorrect actions, respectively, and aim to maximize cumulative rewards. Reinforcement learning algorithms have three key features: they do not require an initial dataset since they learn through action-based rewards and penalties; UEs can act as independent agents, making their own decisions and thereby reducing the workload on the MBS; and they can adapt to dynamic environments with large state and action spaces by updating their strategies based on new experiences.

ML is a collection of methods that allows computers to learn, automate, and optimize a model that helps patterns. [7] It comprises two phases namely, the training

phase and the decision phase. The ML algorithms can be categorized as supervised, unsupervised, and RL. [8]

RL is a type of ML where agents learn by trial and error to make the correct sequential decisions in an environment. The Agent is given positive and negative numerical rewards when it makes correct and wrong actions respectively. The agent tries to make decisions that maximize the cumulative rewards obtained by it. The three distinguishing features of RL algorithms are that they do not require a dataset as an input as the algorithm learns by experimenting with actions in terms of rewards and penalties. Secondly, The UE can act as an independent agent and can make decisions independently. Therefore, the workload of the MBS will be reduced. Thirdly, it can adapt to a dynamic environment with large state and action spaces as it updates its association strategies based on new experiences. Thus, RL algorithms are suitable to perform LB in HetNets.

The paper is structured as follows. Section 2 provides an overview of the research conducted in this field. Section 3 introduces the network model and the proposed LB algorithms. Section 4 examines the network parameters, experimental results and their analysis. Section 5 concludes the paper and outlines potential areas for future research.

2 Literature Survey

A comprehensive review of various literature has been conducted, exploring the present status of the field.

In [9], the authors developed a two-layered LB solution for cellular networks using RL. In the first layer, a centralized agent collects the latest handover statistics and predicts the number of users that will connect to each cell in the next time frame. The second layer involves decentralized agents located at each base station, which use these predictions to adjust Cell Individual Offset (CIO) values, optimizing handover decisions. This approach is compared with traditional decentralized systems and those with constant CIO values, demonstrating superior performance in user distribution and network load management. The system model emphasizes a two-layered approach that combines the benefits of both centralized and decentralized systems, enabling better load predictions and more effective local optimizations. The results show a significant improvement in network performance, particularly in terms of reducing congestion and improving user experience. The authors conclude that their method offers a practical and efficient solution for LB in modern cellular networks, suggesting future work to explore additional reward functions and larger network scenarios.

In [10], the author aims to maximize the number of mobile users offloaded from the Macro Cell (MC) to Small Cells (SCs) through Device-to-Device (D2D) communication in a balanced manner. This involves offloading congested MC users to less loaded SCs, utilizing idle users as relays, and selecting the appropriate users for offloading. The study considers an uplink multi-tier HetNet scenario with a single MC overlaid with multiple SCs connected via backhaul links. When the MC is congested, it offers a fraction of its available resource blocks to SCs for offloading users. The proposed scheme employs dynamic LB based on D2D communications using the Kuhn-Munkres method. The objective is to increase the number of admitted users while ensuring high

LB fairness among small cells. The study addresses challenges related to user-relay link quality, SC capacity, and LB among SCs in a HetNet. The scheme seeks to enhance system performance by optimizing user-relay link quality, SC capacity, and LB among small cells. Performance metrics include maximizing the number of offloaded users, achieving high LB fairness, and improving overall system throughput in a multi-tier HetNet.

In [11], the authors present a fuzzy logic-based LB algorithm for HetNets comprising LTE-A macro cells and 5G small cells, addressing issues arising from different cell types and their load imbalances. It employs a two-stage cell selection process: first, creating a restricted neighbor cell list (NCL) of cells below a threshold load, and then selecting the target cell with the highest RSRP. Simulations show that the algorithm outperforms existing methods in terms of LB, throughput, and call dropping ratio (CDR) across various user speeds. The fuzzy logic system's rules and membership functions optimize HOM values, leading to significant performance gains. Future work suggests enhancing the algorithm with mobility robustness optimization and incorporating additional performance metrics.

In [12], the authors introduce the RL framework to optimize LB in LTE cellular networks. It addresses the issue of traffic imbalance due to the uneven geographical distribution of mobile subscribers. The RL framework adjusts neighbor cell relational parameters, specifically the CIO, to distribute traffic more evenly among cells. The proposed model uses a reward function based on throughput to enhance system performance, achieving up to a 50% improvement in system throughput. Simulations using the NS-3 simulator demonstrate that the RL-based optimization significantly increases overall throughput and reduces the load imbalance among cells. The results highlight the effectiveness of RL in dynamically optimizing network parameters for better resource utilization. Future work suggests exploring different reward functions and expanding the model to include mobility and shadowing effects for more practical scenarios.

In [13], the authors devised a centralized and semi-distributed algorithm for UALB in 5G HetNet using multi-armed bandit (MAB) RL techniques. These algorithms ensure optimal user association and LB among BSs by leveraging a central load balancer (CLB) for real-time decisions. In the centralized approach, the CLB assigns BS connections based on a reward system that updates with each time step, while the semi-distributed approach allows UE to propose actions based on local information and communicate with BSs for rewards. Numerical results indicate that these MAB-based algorithms converge quickly and achieve performance close to a near-optimal benchmark solution, making them suitable for highly dynamic mmWave-enabled networks. The proposed methods can adapt to network dynamics and demonstrate efficiency in dense network environments by maintaining high spectral efficiency and low latency. Additionally, the algorithms are shown to scale effectively with network size and maintain robust performance under varying conditions. The findings highlight the potential of MAB techniques in enhancing the performance and adaptability of future cellular networks.

In [14], the authors investigate the application of Deep reinforcement learning (DRL) for addressing various challenges in mobile 5G and beyond networks. It highlights the necessity for advanced solutions due to the rapid increase in wireless data services and the diverse requirements of 5G applications, such as reduced latency and improved energy efficiency. DRL, which combines RL and deep learning, is proposed as a viable approach for managing dynamic and complex network environments. The paper explains how DRL can optimize decision-making processes in scenarios like power control, offloading, and edge caching, outperforming traditional RL methods by using deep neural networks to approximate the Q-function and handle large state spaces. Case studies and numerical results demonstrate the effectiveness of DRL in achieving superior performance metrics compared to baseline solutions. The authors emphasize that while DRL shows promise, its full potential in 5G and future 6G networks remains largely unexplored, suggesting future research directions in multi-agent and distributed DRL to better handle the decentralized nature of upcoming networks.

In [15], the authors presents a multi-agent reinforcement learning (MARL) approach to address capacity sharing in multi-tenant 5G heterogeneous networks. It employs a Deep Q-Network (DQN) where each agent represents a tenant and learns a policy for allocating resources efficiently across multiple cells, ensuring service level agreement (SLA) satisfaction and optimized resource utilization. The study finds that MARL's multi-agent setup improves scalability and learning efficiency over single-agent approaches, as agents can be added or removed without restructuring the learning model. The proposed approach enables each agent to generalize its policy, allowing a new tenant to adopt an already trained policy without additional training. The policy also dynamically adapts to variations in tenant load demands and spatial heterogeneity within cells, ensuring balanced load distribution. Performance evaluations demonstrate the method's effectiveness in managing SLA satisfaction and resource utilization across dynamic 5G environments.

In [16], the authors introduce an Approximate Q-learning (AQL) method to address load balancing and user association in 5G mobile edge-cloud (MEC) environments with a focus on latency-sensitive services. Network slicing is employed to allocate isolated resources for different applications, enhancing efficiency while managing demands. Given that traditional Q-learning struggles with the vast and dynamic state spaces of 5G networks, this paper modifies the algorithm to handle large state-action spaces efficiently. The AQL model selectively accepts service requests based on revenue generation and delay sensitivity, dynamically adjusting resource allocations. Simulations demonstrate that AQL outperforms standard Q-learning, DQN, and random selection methods across multiple metrics such as resource utilization, slice acceptance, and penalty minimization. Overall, the AQL model enables improved revenue and efficient resource management in 5G MEC by leveraging reinforcement learning to meet stringent application requirements effectively.

In [17], the authors address the critical issue of LB in cellular networks, particularly focusing on idle mode users to enhance network performance and quality of service. Traditional LB methods are rule-based and struggle with adaptability to dynamic traffic changes. To overcome this, the authors propose a novel LB framework based

on DRL, specifically utilizing the soft actor-critic (SAC) algorithm. The main innovation lies in incorporating transfer learning to improve data efficiency, a significant challenge in RL applications. The proposed Transfer Reinforcement Learning-based LB (TRLB) algorithm reuses policies learned from other tasks, thereby reducing the need for extensive data and training time. Experimental results demonstrate that the TRLB method significantly enhances system performance compared to existing baselines, making it robust to environmental changes. In conclusion, this paper highlights the potential of combining deep reinforcement learning with transfer learning to tackle complex LB problems in communication networks.

In [18], the authors propose a particle swarm optimization (PSO)-based approach for network resource management in 5G environments, particularly addressing the needs of heterogeneous networks with software-defined networking (SDN). It identifies challenges in resource utilization within traditional and static 4G networks and highlights the potential of flexible, dynamic resource allocation in 5G enabled by SDN and network slicing. By simulating tenant networks with varied bandwidth needs, the PSO algorithm optimizes bandwidth distribution among congested and underutilized tenant networks, balancing network load effectively. Through iterative learning and information sharing among particles, PSO improves upon genetic algorithms and fusion methods in resource utilization and response time. This approach dynamically reallocates bandwidth in response to congestion or low utilization, achieving optimal resource use while minimizing adjustment costs. Simulation results confirm PSO's superior performance in resource optimization, making it a practical solution for balancing diverse tenant demands in 5G SDN environments.

In previous studies, the authors constructed a number of system models with limited requirements. These works focused on joint optimization algorithms for user association and resource allocation. They considered the fixed deployment of UEs, analyzed system performance based on model performance, and proposed that their model would be useful in situations with certain parameters. However, none of these studies addressed the UARA problem in the context of user mobility. It is important to note that UE locations may vary, making it inefficient to apply the same channel characteristics to all environmental applications. Consequently, the model created by the authors cannot be universally applied in all scenarios with respect to distance. The proposed work considers a signal model for a three-tier HetNet operating under a log-distance channel model in rural areas. It analyzes and plots the performance of LB after varying network parameters and introducing mobility to UEs. This work serves as a benchmark for evaluating the performance of various intelligent UALB methods.

3 Proposed Methodology

3.1 Network Model

A three-tier HetNet in downlink condition consisting of N_m MBSs, N_p Pico BSs (PBSs), and N_f Femto BSs (FBSs) is considered for LB in this work. The BS are represented by the set $BS = \{mbs_1, \dots, mbs_{N_m}, pbs_1, \dots, pbs_{N_p}, fbs_1, \dots, fbs_{N_f}\}$ and with the indexes $\{1, 2, \dots, N\}$ where the total number of BS is given by $N =$

$N_m + N_f + N_p$. In the Network, the MBSs and PBSs are network deployed, while the FBSs are user deployed and all the BSs are considered to be centre-excited.

A total of L number of UEs in the set $UE = \{1, 2, \dots, L\}$ are considered to be requesting the BSs for connection in the HetNet. A total of K orthogonal channels belonging to the set $Ch = \{1, 2, \dots, k\}$ is available to the UEs to connect to a BS. Also, one or more UEs can utilize a common channel to connect with the BSs.

Consider $i \in UE$ trying to be associated with a $j \in BS$ through channel $k \in Ch$. The following parameters need to be computed for every BS and UE pair (i, j) to perform user association and check its effectiveness.

In the network, each UE can associate with only one BS and channel at any given instant, and this is governed by the conditions given by:

$$\sum_{i \in BS} \mathbf{b}_j^i \leq 1 \quad (1)$$

$$\sum_{k \in Ch} \mathbf{c}_j^k \leq 1 \quad (2)$$

Where, $\mathbf{b}_j^i$ and $\mathbf{c}_j^k$ are the binary BS and channel association vector respectively. If the j^{th} UE associated with i^{th} BS, then $\mathbf{b}_j^i = 1$ or $\mathbf{b}_j^i = 0$ otherwise. Similarly, if the j^{th} UE utilizes the k^{th} channel, then $\mathbf{c}_j^k = 1$ or $\mathbf{c}_j^k = 0$ otherwise.

3.1.1 Channel Model

Channel models in wireless communication describe the behavior of the environment through which signals propagate from BS to UEs. These models take into account factors like signal attenuation, interference, and noise that characterize signal propagation in a real-time environment. This work considers the log-distance path loss, which accounts for signal attenuation over distance and the fading model, which simulates signal fluctuations due to multipath propagation. The path loss model is used to characterize the Rural, Urban Micro (UMi) and Urban Macro (UMa) environments. The path loss equations for 5G defined in the 3GPP 38.901 standards document [19] have been used. The path loss equation differs based on the path loss exponent for different wireless environments. The path loss exponent is lower for BSs experiencing less interference and absorption of the transmitted signal. The path loss exponents for the environments under consideration have been determined empirically [20].

The path loss for communication between a BS i and an UE j in rural environment is given by:

$$PL_{i,j}^R (in dB) = 34 + 40 \log_{10}(d) \quad (3)$$

The path loss for communication between a BS i and an UE j in UMa environment is given by:

$$PL_{i,j}^{UMa} (in dB) = 18.3 + 28 \log_{10}(d) \quad (4)$$

The path loss for communication between a BS i and an UE j in UMi environment is given by:

$$PL_{i,j}^{UMi} (in dB) = 37.7 + 20 \log_{10}(d) \quad (5)$$

Where, d is the 2D distance between BS i and UE j given by:

$$d = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2} \quad (6)$$

3.1.2 Signal-to-Interference-plus-Noise Ratio

Signal-to-Interference-plus-Noise Ratio (SINR) represents the ratio of the desired signal power to the combined interference and background noise. A higher SINR indicates a more favorable signal environment, leading to better communication performance. SINR is vital for assessing the reliability and efficiency of communication links. In this work, SINR plays a crucial role in determining the suitability of a UE and BS pair and offloading a UE from a BS. The general formula of $SINR$ between BS i and UE j through channel k is given by:

$$SINR_{i,j}^k = \frac{b_j^i h_i^{j,k} c_j^k p_{i,j}^k}{\sum_{l \in BS \setminus \{i\}} b_j^l h_l^{j,k} c_j^k p_{l,j}^k + BW \cdot N_o} \quad (7)$$

Where, $h_i^{j,k}$ is the channel gain derived from $PL_{i,j}$ defined in section 3.1.1, $p_{i,j}^k$ is the power transmitted by BS i , BW is the channel bandwidth and N_o is the noise power density.

3.1.3 Transmission Rate

The transmission rate is the speed at which digital data is transferred between a UE and BS. It can also be defined as the amount of data transmitted in a given time between a UE and BS. In this work, the transmission rate is an important indicator for representing the efficiency of a communication link as because it measures the speed of data how quickly data can be exchanged over a channel.

According to the Shannon capacity theorem, the maximum transmission rate $r_{i,j}$ for a link between BS i and UE j is less than or equal to the capacity $C_{i,j}$ of the link. The capacity $C_{i,j}$ is given by:

$$C_{i,j}^k = BW \log_2(1 + SINR_{i,j}^k) \quad (8)$$

In this work, the maximum possible transmission rate is taken into consideration. So, the $r_{i,j}$ is given by:

$$r_{i,j}^k (in \text{ bps}) = BW \log_2(1 + SINR_{i,j}^k) \quad (9)$$

For the whole network, the average transmission rate $\bar{R}$ is given by:

$$\bar{R} (in \text{ bps}) = \frac{\sum_{i \in UE} r_{i,j}^k}{L} \quad (10)$$

3.1.4 Mobility Model

Mobility is incorporated into the UE to introduce dynamism to the system. In this study, the Random Walk mobility model is employed to simulate UE movement. This

model operates on a stochastic process, where UEs change direction randomly at each time step [21]. The updated coordinates after each step of UE mobility are given by:

$$x' = x + l \cos(\theta) \quad (11)$$

$$y' = y + l \sin(\theta) \quad (12)$$

Where, (x, y) is the initial coordinates, l is the step size and $\theta \in (0, 2\pi)$ is an angle chosen randomly that controls the mobility direction at each step.

3.2 User Associated Load Balancing Algorithms

3.2.1 MaxSINR Algorithm

The MaxSINR approach follows a naive method of performing user association and load balancing in HetNets. This algorithm assigns each UE to a single channel, which limits its ability to accommodate multiple UEs within the same channel. Consequently, this restriction results in diminished bandwidth availability and lower transmission rates for UEs. This leads to reduced bandwidths and transmission rates.

The MaxSINR algorithm initially associates all the UEs with the MBS_1 and the corresponding $SINR_{1,j}$ for every $j \in UE$ is determined. Then PBSs and FBSs within the range of each UEs are determined and the corresponding SINR values for those associations are computed. If the $SINR_{1,j}$ is less than the threshold of 10 dB, then the user association is changed to a PBS or FBS offering maximal $SINR$ and the corresponding transmission rate is computed. This process is repeated until all the UEs identifies and associates with a BS offering the highest $SINR$ and transmission rates.

3.2.2 Q-Learning Algorithm

Q-learning (QL) is a model-free RL algorithm designed to find the optimal action-selection policy in a given environment. Unlike some other RL methods, the QL algorithm does not require a model of the environment, making it highly adaptable to various situations. Additionally, QL algorithm does not require a dataset for training the ML model as the learning happens on the go. Thus, QL is well-suited for performing UALB in 5G HetNets.

A QL algorithm involves an agent making decisions within an environment and receiving rewards based on those decisions. These rewards can be positive for successful actions and negative for unsuccessful ones. Objective of the agent is to take actions that maximize positive rewards over time. In the context of LB in 5G HetNets, the agent represents a UE i , the environment is the 5G HetNet, and the action involves UE i selecting a BS j and channel k for association. Positive rewards are received when the chosen BS and channel pair establish a successful connection.

The QL algorithm aims to learn the Q-function, which estimates the expected utility or the total accumulated reward of choosing a BS and channel pair in a given state and following the optimal policy thereafter. The Q-function, denoted as $Q_i(s, a_i)$, represents the value of action a_i of a particular UE i picking a BS and channel pair in

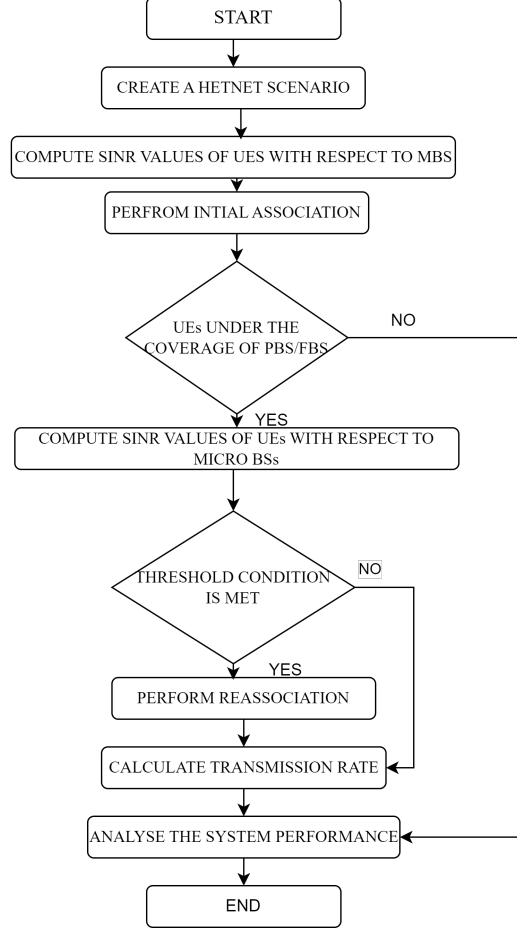


Fig. 2 Working Mechanism of MaxSINR based User Association

the current network state s . The goal of the algorithm is to learn the optimal Q-values, $Q^*(s, a)$ which satisfy the Bellman equation:

$$Q^*(s, a) = \mathbb{E}[r_t + \gamma \max_{a'} Q^*(s_{t+1}, a') | s_t = s, a_t = a] \quad (13)$$

where r_t is the reward received after taking action a in state s , and s_{t+1} is the state resulting from this action. The discount factor γ ($0 \leq \gamma \leq 1$) determines the importance of future rewards, with values closer to 1 making future rewards more significant.

The steps involved in QL algorithm is illustrated in Figure 3. Initially, the Q-function of all the UEs $Q_i(s, a)$ is set to zero. Using the ϵ -greedy approach [22], the algorithm either chooses to explore by gaining more information about the environment (picks a BS and channel pair randomly) or exploit by leveraging the current knowledge of the environment to select actions that are known to be positive (picks BS and channel based on Q-function). Then a connection request is sent by UE and

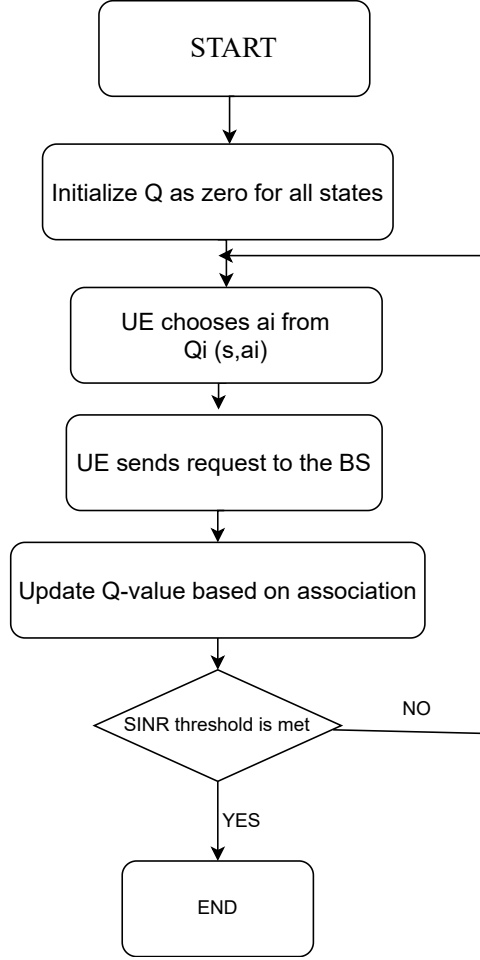


Fig. 3 Working Mechanism of QL Algorithm

a reward is observed based on the success of the connection. Then, the Q-function is updated based on association success using:

$$Q(s, a) \leftarrow Q(s, a) + \alpha(R(s, a) + \gamma \max_{a'} Q(s', a') - Q(s, a)) \quad (14)$$

where α ($0 \leq \alpha \leq 1$) is the learning rate that controls how much new information overrides the old information. This equation ensures that the Q-values are progressively refined based on the observed rewards and the estimated value of the subsequent states. The above process is repeated until all the UEs associate with a BS successfully. Q-learning is guaranteed to converge to the optimal Q-functions, $Q(s, a)$, as long as all state-action pairs continue to be updated and the learning rate α decreases appropriately over time.

3.2.3 Deep Q-Learning

Deep Q-Learning (DQL) is an extension of Q-learning that leverages deep neural networks to approximate the Q-function instead of the Bellman-Ford equation [14]. DQL makes it possible to handle high-dimensional state spaces commonly encountered in real-world reinforcement learning problems [23]. Instead of using a Q-table, DQL employs a neural network to approximate the Q-values for each action in a given state. An experience replay memory is used to store the previous state transitions and actions. Using the experience replay memory DQL identifies correlations between consecutive experiences and sample batches for training the neural network. The flow chart in Figure 4 shows the steps involved in DQL.

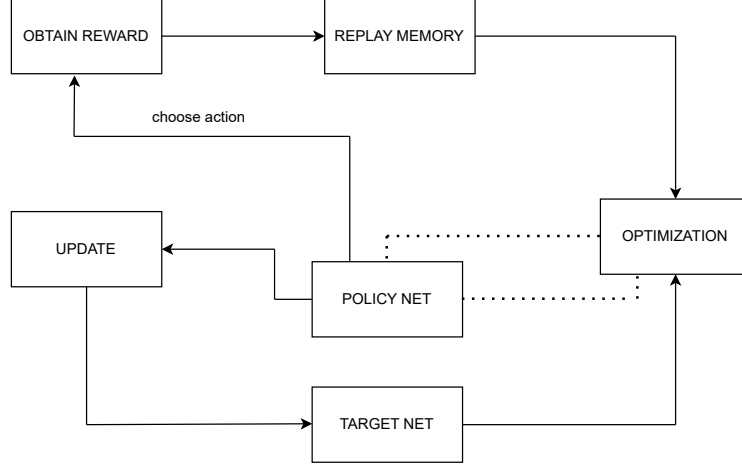


Fig. 4 Working Mechanism of DQL Algorithm

Initially, a policy and target deep neural network, represented by θ and θ' , both comprising an input layer (N neurons), two hidden layers (62 and 32 neurons) and output layers (LK neurons) is initialized with random weights. Using the ϵ -greedy approach [22], the algorithm either chooses to explore by gaining more information about the environment (picks a BS and channel pair randomly) or exploit by leveraging the current knowledge of the environment to select actions that are known to be positive (picks BS and channel based on Q-Values). After associating the UE with the chosen BS and channel pair, a positive and negative reward is given to the algorithm for a successful and unsuccessful connection respectively. This previous state and rewards are stored in the replay memory. Finally, a mini-batch from the replay memory is sampled and gradient descent step is executed to update policy network θ . Periodically after certain iterations, the target network θ' is updated to match the policy network θ .

4 Results

4.1 Simulation Parameters

A HetNet consisting of 1 MBs, 2 PBs and 4 FBs along with 30 UEs is simulated to study the performance of the three proposed algorithms for UALB. The other parameters considered for simulation are given in Table 1. The proposed work considers the combination of two different micro BSs like PBS and FBS and considers dynamic parameters like mobility. Additionally, the UALB is performed for three different channel models.

Table 1 Network Simulation Parameters

Parameter	Value
Channel Bandwidth W	180 KHz
Centre Frequency f	2 GHz
Number of UEs L	30
Number of Channels	25
Number of BSs	MBS-1, PBS-2, FBS-4
Transmit Power of BSs	MBS-40, PBS-30, FBS-8 dBm
BS Radius of Coverage	MBS-500, PBS-200, FBS-100 m
Noise Power Density N_o	-174 dB

The HetNet under simulation is hexagonal cellular network of radius 500m. The MBS is placed in the centre of network at (500,500). The locations of other UEs and BSs relative to the MBS are listed in Table 2 and Table 3 respectively. The overall network under simulation is depicted in Figure 5.

Table 2 Location of UEs

UE	X-coordinate (in m)	Y-coordinate (in m)
UE1	791.44	406.75
UE2	151.81	251.48
UE3	609.99	664.04
UE4	580.36	410.20
UE5	399.05	125.72
UE6	570.46	486.52
UE7	606.94	943.48
UE8	586.63	694.81
UE9	227.54	457.27
UE10	415.82	817.65
UE11	750.26	800.56
UE12	319.55	701.18
UE13	574.02	170.55
UE14	277.47	480.04
UE15	868.57	610.73
UE16	686.63	842.14
UE17	621.01	459.74
UE18	676.26	41.37
UE19	725.60	658.92
UE20	115.31	651.88
UE21	325.41	505.29
UE22	578.07	449.98
UE23	367.13	283.11
UE24	223.06	464.89
UE25	647.14	839.16
UE26	577.42	13.79
UE27	882.62	202.55
UE28	840.97	317.75
UE29	549.52	640.25
UE30	451.53	594.64

Table 3 Location of BSs

BS	X-coordinate (in m)	Y-coordinate (in m)
MBS	500	500
PBS1	500	800
PBS2	500	200
FBS1	600	600
FBS2	350	500
FBS3	200	170
FBS4	800	-290

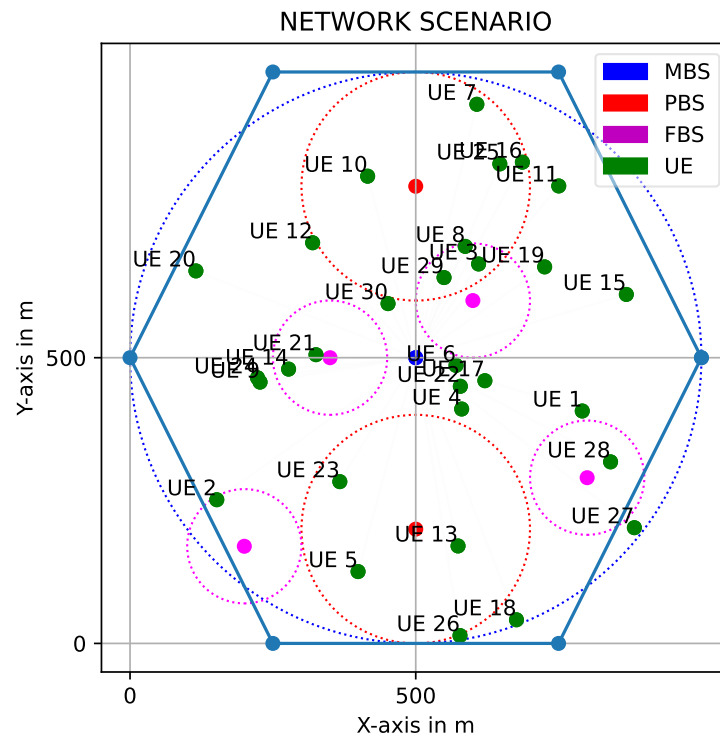


Fig. 5 Network Scenario

4.2 Transmission Rate Analysis

Initially, all UEs are associated with the MBS and then offloaded and reassociated using the proposed algorithms. The transmission rates of all 30 UEs from the simulation, both before and after LB using the three proposed algorithms across various channel models, are tabulated in Table 4.

Table 4 Your Table Caption

UE	Before UALB										After UALB									
						Max-SINR					Q-Learning					Deep Q-Learning				
	Rural		UMi		UMa	Rural		Urban Micro		Urban Macro	Rural		Urban Micro		Urban Macro	Rural		Urban Micro		Urban Macro
	Rate	BS	Rate	BS	Rate	Rate	BS	Rate	BS	Rate	Rate	BS	Rate	BS	Rate	Rate	BS	Rate	BS	Rate
1	0.66	M	0.56	P1	0.61	1.32	M	0.79	P1	1.19	4.40	M	1.68	M	3.98	4.43	M	1.68	M	4.40
2	1.00	M	0.69	M	0.92	0.86	M	0.69	M	1.05	4.05	M	1.37	M	3.78	3.84	F3	1.37	F3	4.05
3	0.61	F3	0.55	F3	0.57	1.53	F3	0.98	F1	1.39	4.72	F1	0.87	F1	3.71	4.65	F1	3.00	P1	4.21
4	1.02	F1	0.70	F1	0.94	1.50	F1	0.94	P1	1.20	M	M	2.64	M	4.54	M	M	2.64	M	5.08
5	0.72	F4	0.57	F4	0.66	1.46	F4	0.89	F4	1.32	M	M	2.00	M	3.84	M	P2	0.60	M	4.23
6	0.70	F3	0.57	F3	0.65	1.37	F3	0.75	F3	1.23	M	M	3.18	M	4.85	M	M	3.18	M	5.45
7	1.06	F2	0.72	F2	0.98	1.51	F2	0.96	F2	1.36	M	M	1.64	M	3.74	P1	P1	1.64	M	4.11
8	1.04	P2	0.71	P2	0.96	1.29	P2	0.76	P2	1.15	4.05	F3	0.66	F1	3.48	M	F1	0.66	F1	4.05
9	0.83	P2	0.61	P2	0.76	1.33	P2	0.78	P2	1.19	M	M	1.78	M	4.04	P1	M	1.78	M	4.47
10	1.00	M	0.69	M	0.92	1.20	M	0.69	M	1.05	M	M	2.40	M	3.94	M	P1	0.81	P1	0.43
11	0.95	M	0.66	P1	0.88	1.53	P1	0.91	P1	1.36	M	M	1.42	M	3.83	M	M	1.42	M	4.22
12	0.94	M	0.66	M	0.86	1.14	M	0.66	M	0.99	M	M	1.80	M	4.06	P2	M	1.80	M	4.49
13	0.67	P2	0.56	P2	0.62	1.27	P2	0.75	P2	1.13	M	M	0.91	M	3.92	M	P2	0.91	P2	0.49
14	0.88	P2	0.63	P2	0.80	1.34	P2	0.79	P2	1.20	F3	M	0.89	M	4.17	M	F2	0.89	M	4.63
15	0.70	P2	0.57	P2	0.65	1.19	P2	0.71	P2	1.05	M	M	1.44	M	3.84	M	M	1.44	M	4.23
16	0.93	F4	0.65	F4	0.85	1.14	M	0.65	F4	1.05	4.14	P1	1.43	M	3.84	M	M	1.43	M	4.22
17	1.30	M	0.87	M	1.20	1.30	M	0.87	M	1.18	M	M	2.58	M	4.5	M	M	2.58	M	5.03
18	1.14	F2	0.77	F2	1.05	1.20	M	0.77	F2	1.11	M	M	1.19	M	3.7	M	M	1.19	M	4.05
19	1.01	P2	0.70	P2	0.94	1.16	P2	0.81	P2	1.07	4.47	M	1.78	M	4.04	M	P1	1.78	M	4.47
20	0.85	M	0.62	M	0.78	1.26	F4	0.89	F4	1.17	4.18	M	1.36	M	3.8	F2	F1	1.36	M	4.18
21	1.13	M	0.76	M	1.04	1.13	M	0.76	M	1.02	M	M	1.48	F2	4.28	M	F2	1.48	F2	5.02
22	1.19	F2	0.81	F2	1.11	1.11	M	0.81	F2	1.03	M	M	2.92	M	4.69	P2	M	2.92	M	5.27
23	0.85	M	0.62	M	0.78	1.05	M	0.62	M	0.76	4.29	P2	1.77	P2	3.78	M	P2	1.77	M	4.53
24	1.15	F4	0.78	F4	1.07	0.85	M	0.78	M	1.05	4.46	M	1.77	M	4.04	M	M	1.77	M	4.46
25	0.83	P2	0.61	P2	0.76	1.31	P2	0.90	P2	1.20	4.26	P1	1.80	M	3.87	M	P1	0.39	M	4.26
26	0.98	M	0.68	M	0.90	0.83	M	0.68	M	0.88	M	M	1.18	M	3.7	M	M	1.18	M	4.05
27	0.57	M	0.54	M	0.54	0.98	M	0.54	M	0.52	4.06	M	1.20	M	3.71	F4	M	1.20	M	4.06
28	0.73	P2	0.58	P2	0.67	1.08	M	0.58	M	0.99	M	F4	1.77	F4	3.87	F1	F6	3.21	F4	0.37
29	0.91	M	0.64	M	0.83	1.07	M	0.64	P1	0.99	4.33	F1	1.30	M	4.41	M	M	0.60	F1	4.33
30	1.09	M	0.65	P1	1.01	1.12	M	0.76	P1	1.03	5.17	M	2.77	M	4.61	M	M	2.77	M	5.17

The average transmission rate of the network under different channel models and UALB algorithms is determined, and presented in Table 5 and Figure 6. Following LB with MaxSINR, QL, and DQL algorithms, an average increase of 34%, 387% and 392% in transmission rates was observed in the rural environment. Similarly, in the UMa environment, increases of 30%, 376% and 382% were noted with the same algorithms. In the UMi environment, LB with MaxSINR, QL, and DQL algorithms resulted in increases of 317%, 158%, and 151% in transmission rates respectively.

Table 5 Average Transmission Rate before and after UALB

Algorithm	Channel Model	Average Transmission Rate (Mbps)	
Before UALB	Rural	0.91	
	Urban Micro	0.66	
	Urban Macro	0.84	
After UALB	Max-SINR	Rural	1.22
		Urban Micro	0.77
		Urban Macro	1.1
	QL	Rural	4.44
		Urban Micro	1.7
		Urban Macro	4.02
	DQL	Rural	4.48
		Urban Micro	1.65
		Urban Macro	4.07

The transmission rate is at its lowest prior to LB when all UEs are connected to the MBS. The implementation of the MaxSINR approach for LB redistributes some UEs from the MBS to micro BSs. However, there is only a marginal improvement in the transmission rate post LB, as multiple UEs served by a single BS share a single channel, leading to increased interference and decreased transmission rates. On the other hand, RL algorithms possess the capability to assign both a BS and channel to each UE. Thus, QL and DQL algorithms exhibit a significant enhancement in the average transmission rate following LB.

From the above analysis, it is noted that the rural channel model exhibits the highest transmission rate, followed by the UMa and UMi channel models. The rural channel model assumes environments with sparse buildings and population, resulting in minimal interference of transmitted signals and consequently, high transmission rates. Conversely, the UMi channel model depicts densely populated cities with significant interference of transmitted signals, leading to lower transmission rates.

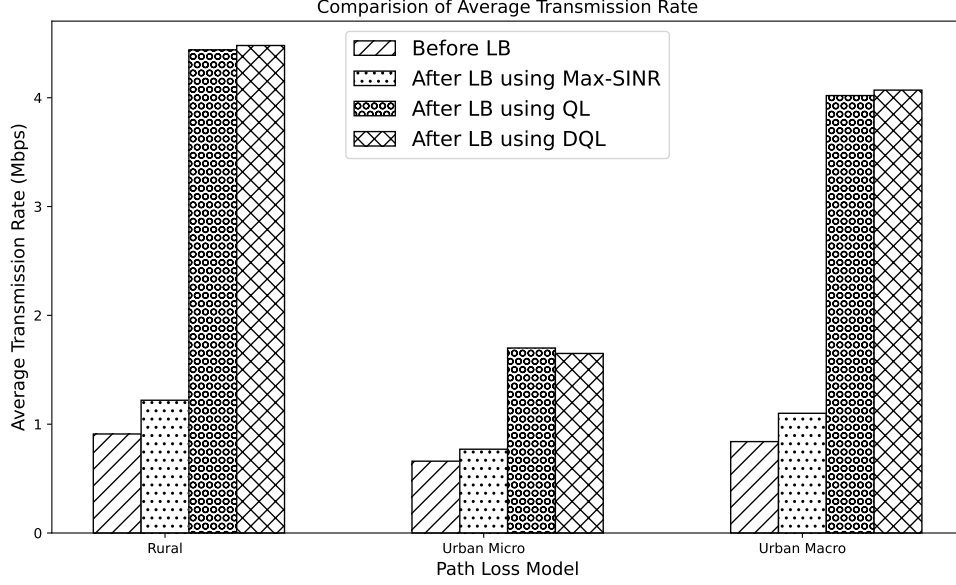


Fig. 6 Comparison of Average Transmission Rate at different network scenarios

4.3 UE Scalability Analysis

The scalability of the proposed algorithm is verified by increasing the number of UE from 15 to 45. The comparison of transmission rate between the QL model and the Deep QL model for varied UEs is shown in figure 7.

The figure 7 suggests that the performance of the algorithm remains unaffected by the rising number of UEs. The Q-learning algorithm can handle over 100 UEs, while the Deep QL algorithm can manage up to 45 users, owing to its considerable computational complexity.

4.4 Computational Complexity Analysis

The time and space complexity of the three proposed algorithms are derived in terms of number of UEs L , BSs N , channels K and replay memory size R and listed in table 6.

Table 6 Computational Complexity Analysis

	Max-SINR	Q-Learning	Deep Q-Learning
Time Complexity	$O(L)$	$O(LNK)$	$O(LN^2K^2)$
Space Complexity	$O(L)$	$O(LNK)$	$O(LN^2K^2R)$

It can be observed that the MaxSINR algorithm is the least computationally expensive in terms of both runtime and space. The DQL needs high computation power due

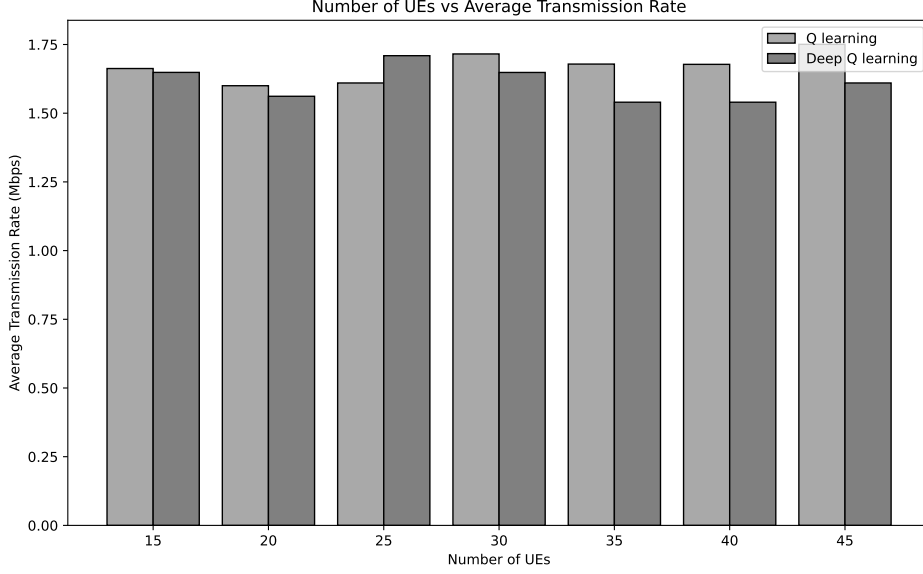


Fig. 7 Transmission rate for varying numbers of UES

to the involvement of neural networks that requires more calculations and memory. The QL algorithm is considered to be moderately computationally expensive among the three proposed algorithms.

The plot in figure 8 shows the change total time taken by QL and DQL to execute on an Windows i5 system with 8GB RAM on scaling the total number of UEs in the network. The gradual increase in the number of UEs results in an exponential increase in simulation time as predicted by the computational analysis detailed in table 6.

4.5 Convergence Analysis

Convergence analysis is the evaluation of how an algorithm or a procedure reaches a stable and consistent state over time or through iterations. By processing training data and adjusting model parameters, it analyses whether the algorithm converges to a stable solution. It is used to evaluate the behavior of learning algorithms.

It is inferred from Figure 9 that both QL and Deep QL algorithms begin to converge from the 12th Episode. Before the 12th episode, the algorithms perform associations on a trial-and-error basis and gain rewards. Based on the rewards, the algorithm learns and begins to converge. The convergence of the QL algorithm is more stable than the Deep QL algorithm because it requires fewer samples to converge compared to DQL and it uses neural networks to approximate the Q-values, which introduces issues like function approximation errors.

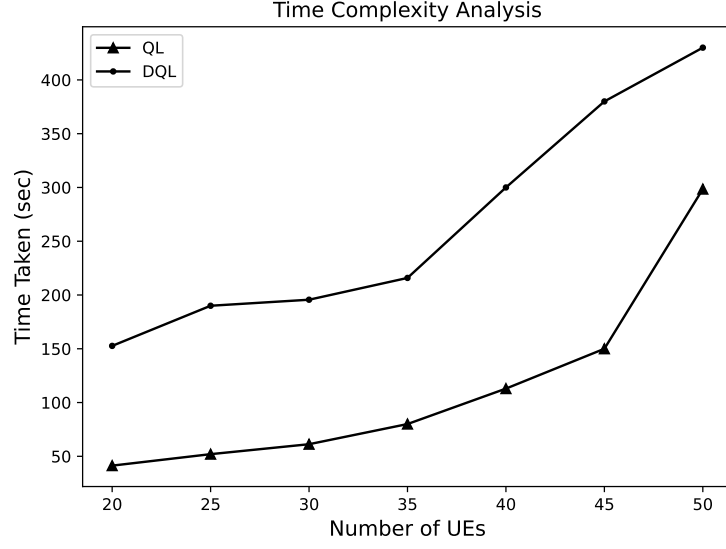


Fig. 8 Simulation Runtime Analysis

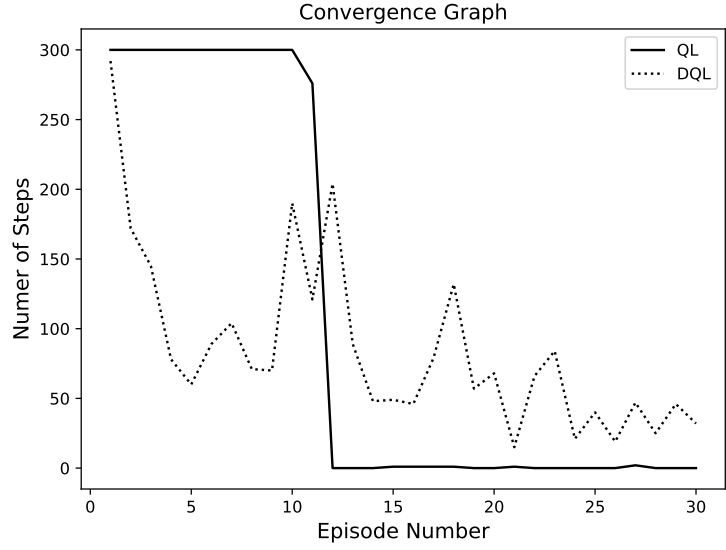


Fig. 9 Convergence Plot for QL and Deep QL Algorithms

4.6 Mobility Analysis

Mobility is integrated into the UE to introduce dynamism to the system. In this study, the Random Walk mobility model is used to simulate UE movement. This model functions based on a stochastic process, where UEs randomly change direction at

each time step. The parameters listed in the Table 1 can be adjusted to control UE movements, allowing for fine-tuning of the UE's behavior.

Figure 10 displays the transmission rate of the system as the percentage of mobile users varies. The plot suggests that mobility has minimal impact on the average transmission rate. Thus, the proposed algorithms demonstrate robustness in dynamic environments.

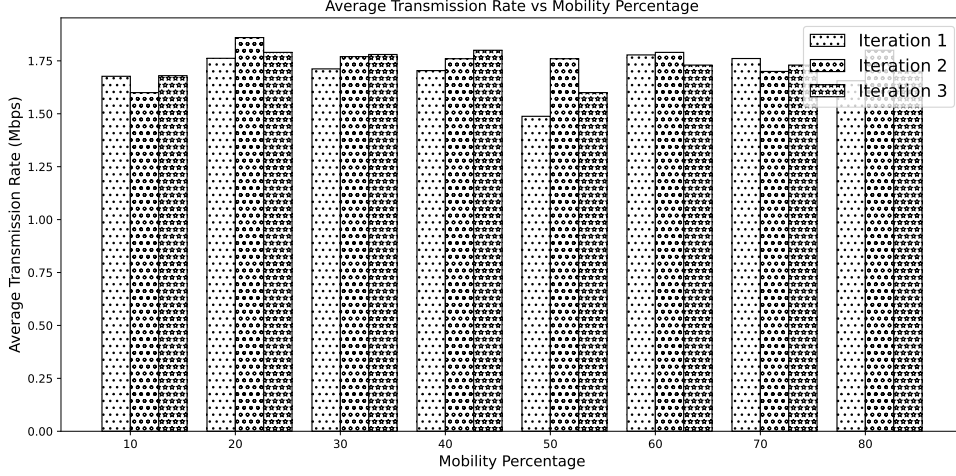


Fig. 10 Transmission rate for varying Mobility in Network

4.7 Comparison of Algorithms

The three algorithms are compared based on the study's results, and the comparison is presented in a Table 7.

Table 7 Comparison of Algorithms

S. No.	Feature	Max-SINR	Q-Learning	Deep Q-Learning
1	Algorithm Type	Brute Force	Reinforcement Learning	Reinforcement Learning
2	Transmission Rate	Low	High	High
3	Maximum UE capacity	>100	>100	<50
4	Convergence	Stable	Stable	Less Stable
5	Computational Complexity	Less	Medium	High
6	Adaptable to Mobility	Yes	Yes	Yes

5 Conclusion and Future Work

The design and simulation of a downlink three-tier 5G HetNet consisting of 30 UEs, 1 MBS, 2 PBSs, and 4 FBSs is performed. LB is performed using MaxSINR, QL, and DQL algorithms and these algorithms are compared based on average transmission rate, computational complexity, and convergence. Deep Q-learning demonstrates enhanced transmission rates, while Q-learning shows quicker convergence and reduced computational complexity. The convergence of Deep Q-Learning algorithms can be improved by using deeper neural networks, but this approach will require more computational power to implement. Additionally, the proposed algorithms exhibit robustness across various network sizes, channel models, and mobility scenarios. Also, the minimum transmission rate required for eMBB services such as audio and video calls is achieved by the proposed algorithm. It is observed that the rural scenario outperforms other network scenarios because it experiences lower interference levels. In this setup, the sparse distribution of buildings and population results in fewer obstacles, reducing signal interference and allowing higher transmission rates. This makes rural models particularly favourable compared to urban micro and urban macro settings, where high population density and structural interference lower transmission efficiency. The proposed algorithms have the ability to allocate a channel and BS pair to the UEs which is said to be a resource block. Future research can enhance transmission rates for supporting eMBB services such as high definition video streaming by assigning multiple resource blocks to a single UE. Additionally, future work could concentrate on evaluating the proposed system using channel models simulating real-world scenarios with high UE density, such as stadiums, where effective LB is crucial.

Acknowledgements. The authors wish to extend their sincere thanks to the Centre for Internet of Things (CIoT) at the Madras Institute of Technology Campus, Anna University, for providing the laboratory facilities essential for this research.

Declarations

Conflict of Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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